

# The Word Garden: Design, Development, and Expert Evaluation of a Multimodal Multiplayer Literacy Game for Children with Dyslexia

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**Abstract**—Children with dyslexia face not only literacy challenges but also heightened anxiety and diminished motivation from repeated failure. Existing digital interventions prioritize phonological drills while under-addressing emotional safety and social support, operating predominantly as single-user applications. This paper presents The Word Garden, a cooperative multiplayer educational game employing dual-device architecture to integrate emotionally safe social interaction with literacy skill development for children aged 8-12 with dyslexia. The system separates collaborative PC-based exploration from private smartphone-based multisensory literacy tasks, enabling children to practice challenging exercises without peer visibility while maintaining social connection through voice communication. The design emphasizes cooperative mechanics without competitive elements, low-visibility interaction, and non-punitive feedback to reduce performance anxiety. Expert validation by consultant child and adolescent psychiatrists confirmed alignment between design features and therapeutic principles for literacy intervention. Technical evaluation demonstrated system feasibility with stable multiplayer synchronization, 98.3% event delivery accuracy, and zero crashes across 48 hours of testing with 2-5 concurrent players. This work constitutes Phase 1 of a longitudinal study; child user evaluation and empirical learning outcomes will be addressed in Phase 2. This formative evaluation establishes technical and design foundations for investigating whether cooperative multiplayer environments can effectively support both literacy development and social-emotional well-being in children with dyslexia.

**Keywords**—AI-assisted literacy learning, Dyslexia intervention, Educational game design, Multisensory learning, Smartphone-based learning

## I. INTRODUCTION

Dyslexia, a neuro-developmental disorder affecting 5-10% of children globally [1], is characterized by persistent deficits in word recognition, decoding, and spelling despite adequate intelligence and instruction [2]. Beyond academic challenges,

affected children experience heightened anxiety, diminished self-esteem, and reduced reading motivation from repeated failure [3].

Traditional phonics-based interventions demonstrate short-term gains but fail to sustain engagement due to their monotonous, high-pressure nature, often exacerbating fear of errors and low self-competence [1], [3]. Recent advances in immersive technologies have shown potential to modernize education through interactive, risk-free learning environments [4]. Game-based approaches offer particular promise by reducing failure anxiety through playful mechanics and fostering intrinsic motivation [5].

However, existing dyslexia interventions under-address motivation and emotional safety, prioritizing cognitive drills over affective factors [3]. Current digital tools are predominantly single-user applications lacking social support, creating isolated experiences [1]. While game-based learning shows promise for literacy [6], [7], multiplayer literacy games targeting dyslexia remain rare [6]. Existing approaches involve individual play with incidental social interaction rather than networked cooperative gameplay with intentional collaborative mechanics addressing both literacy and social-emotional needs [8].

This paper introduces "The Word Garden," a cooperative multiplayer educational game employing dual-device architecture for dyslexic children, featuring PC-based collaborative exploration and smartphone-based private literacy tasks. This architecture enables cognitive compartmentalization while maintaining emotional safety. Children engage in literacy exercises on personal devices without peer visibility, reducing performance anxiety while preserving cooperative gameplay benefits. The system incorporates: (1) dual-device architecture combining PC multiplayer world-building with smartphone

literacy tasks; (2) cooperative mechanics emphasizing teamwork over competition; (3) multi-sensory literacy tasks (visual, auditory, tactile) within a narrative garden ecosystem; (4) emotional safety principles including error normalization and low-stakes progression. This paper presents the design, development, and preliminary technical and expert evaluation as the foundational phase of a longitudinal study. Child user evaluation, empirical learning outcomes, and usability assessment with the target population will be addressed in Phase 2 of this longitudinal study.

## II. LITERATURE REVIEW

Existing literature on technology-enhanced dyslexia interventions reveals a predominant focus on phonological skill development, with emerging interest in game-based and multisensory approaches. This review synthesizes research across dyslexia interventions, game-based learning, and multisensory methodologies, identifying gaps in cooperative, emotionally supportive literacy interventions.

### A. Dyslexia Interventions and Digital Tools

Phonics-based instruction employing explicit drills in blending, segmenting, and decoding builds phonological awareness, yielding moderate word recognition gains [9], [10]. However, traditional interventions demand intensive, specialist-led sessions that limit scalability [11]. Digital platforms such as GraphoGame, Lexia Core5, and Nessy Learning address scalability through adaptive phonics practice [11], yet remain predominantly single-user applications, isolating learners from peer support [10]. Critically, emotional factors including self-efficacy and failure apprehension receive limited attention [3].

### B. Game-Based Learning and Emotional Safety

Game-based learning supports literacy through immersive interactions and immediate feedback [6], [7]. While systems like GraphoGame demonstrate phonics gains [12], [13], they operate in single-user modes, limiting social reinforcement [7]. Research suggests multiplayer cooperative mechanics amplify outcomes through peer collaboration [12], yet multiplayer literacy games targeting dyslexia remain rare [6]. Studies on social presence indicate peer dialogue reduces isolation [14], [15], while emotional safety mechanisms like non-punitive feedback mitigate performance apprehension [15], [16]. However, existing tools rarely integrate such features within cooperative literacy contexts [14].

### C. Multisensory Learning Approaches

Multisensory learning integrating visual, auditory, and kinesthetic modalities strengthens neural connections for phonological processing [17], [18]. For children with dyslexia, multisensory methods enhance decoding accuracy and memory retention through motor engagement like letter tracing [19], [20]. Current digital implementations remain confined to single-device, tablet-based experiences [17], limiting distributed multisensory interaction in collaborative contexts [18].

### D. Research Gaps

Literature analysis reveals critical gaps across motivational, social, and technological dimensions [3]. First, interventions operate predominantly in single-user formats, lacking cooperative multiplayer implementations for dyslexia [10], [12]. Multiplayer environments could provide peer scaffolding that normalizes challenges and sustains engagement [7], [12]. Second, single-device constraints limit kinesthetic integration, while touch-based tracing on smartphones strengthens letter formation memory [19], [20] and dual-device architectures could separate private practice from collaborative activities. Third, emotional safety features are rarely embedded in dyslexia interventions [15]. The Word Garden addresses these gaps through dual-device multisensory delivery, cooperative multiplayer mechanics, and emotionally supportive interaction patterns.

## III. SYSTEM DESIGN

This section presents The Word Garden's design framework and dual-device architecture, encompassing cooperative gameplay mechanics, emotional safety principles, and technical implementation. Throughout this paper, the terms multisensory and multimodal are used interchangeably to describe learning activities engaging visual, auditory, and tactile modalities simultaneously.

### A. Design Framework

The Word Garden targets children aged 8–12 with dyslexia within supervised educational contexts where trained facilitators monitor sessions, accommodating varied cognitive and emotional needs. The system employs a fantasy-themed narrative featuring interconnected islands accessible through portals, as shown in Figure 1, with an Archangel character serving as in-game mentor. This narrative embeds literacy tasks within exploratory context rather than isolated drills, shifting from evaluative testing to discovery-based learning, in line with Self-Determination Theory principles of autonomy and competence, and Universal Design for Learning's emphasis on multiple means of engagement [5].



Fig. 1. PC game environment showing interconnected islands

Cooperative mechanics prioritize teamwork over competition through shared objectives with progression dependent on collective success. Real-time voice communication enables coordination within a low-visibility environment where players

are not visually exposed to one another, as illustrated in Figure 2, reducing social comparison anxiety [3]. The game excludes competitive elements including leaderboards and rankings, fostering peer scaffolding that normalizes literacy struggles [7].



Fig. 2. Multiplayer cooperative gameplay showing multiple players exploring shared environment

Emotional safety is embedded through voice-only communication without visible avatars, non-punitive feedback treating incorrect attempts as exploratory iterations [15], incremental task complexity, and progress-based rewards rather than comparative achievement [3].

### B. Dual-Device Architecture

The system employs distributed dual-device architecture where the PC manages world rendering, multiplayer synchronization, and narrative progression, while smartphones deliver structured multisensory literacy exercises, as illustrated in Figure 3. The dual-device architecture is motivated by well-documented evidence that children with dyslexia experience significantly elevated social anxiety, particularly fear of peer judgment during reading tasks, compared to typically developing peers [3], [21]. Exposing literacy struggles within a shared visible environment risks amplifying this anxiety; the architectural separation of private smartphone-based practice from shared PC-based exploration directly addresses this tension by enabling individual engagement with challenging tasks without peer visibility. Task-triggering events flow from PC to smartphone and completion signals return to the PC, intentionally separating immersive social interaction from individual learning activities: children practice literacy privately while remaining socially connected, resolving the tension between peer support and performance anxiety [3].

The PC component handles 3D world rendering, environmental interaction, multiplayer session management, and real-time voice communication. When players encounter task-triggering objects such as hidden chests, the PC generates events transmitted to paired smartphone applications. The smartphone component delivers multisensory literacy tasks including letter tracing with stroke order guidance, letter and phoneme recognition, and object-letter association exercises, as shown in Figure 4. Only task completion status (not raw performance data) is transmitted to the server, ensuring

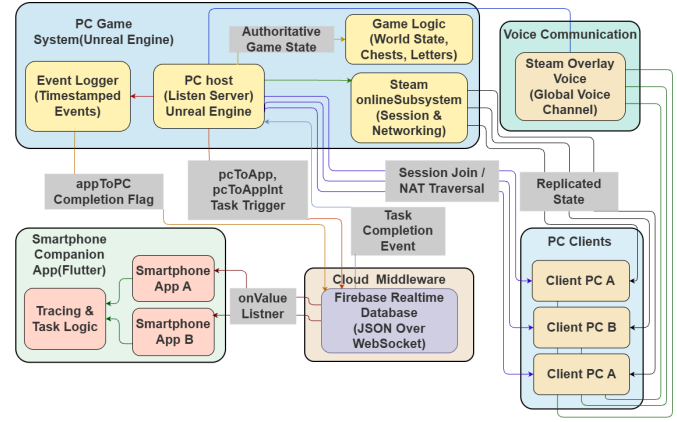


Fig. 3. System architecture showing dual-device communication flow

privacy; future child deployment phases will obtain ethics board approval and informed parental consent [3].

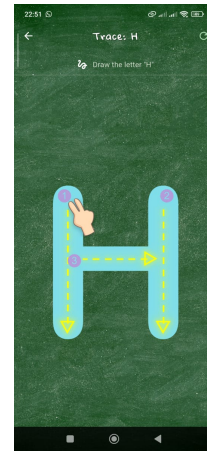


Fig. 4. Smartphone letter tracing interface

### C. Gameplay Flow and Implementation

Players select difficulty level via smartphone and join sessions using four-digit codes generated by the PC host. Gameplay alternates between cooperative exploration on shared PC environment and individual literacy practice, following the sequence illustrated in Figure 5. Upon discovering hidden chests, smartphone notifications initiate literacy tasks. Successful completion unlocks chests, revealing letters added to collective inventory, while failed attempts trigger encouraging feedback without penalties. Players regroup at the central island to collaboratively arrange collected letters into valid words through voice discussion, receiving rewards including stars and badges.

The session-based connection mechanism eliminates persistent user accounts, with session identifiers expiring after gameplay. The modular architecture supports 2-6 concurrent players per session on standard educational network infrastructure, with future extensibility for facilitator monitoring dashboards and adaptive difficulty algorithms.

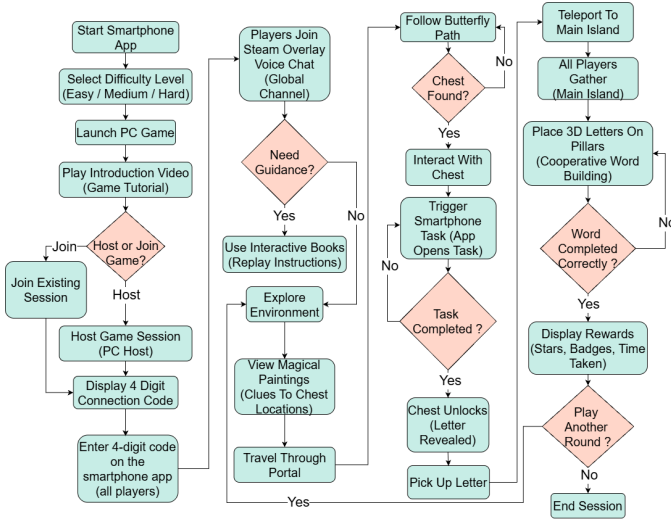


Fig. 5. Gameplay flow diagram showing session initialization, cooperative exploration, literacy tasks, and word formation cycle

## IV. METHODOLOGY

### A. Research Approach

This study employs a formative evaluation methodology comprising iterative system development, technical validation, and expert review. As the foundational phase of a longitudinal investigation, this work establishes technical feasibility and design appropriateness before proceeding to empirical assessment with children. The methodology integrates two complementary evaluation components: controlled technical performance testing and expert validation by consultant child and adolescent psychiatrists, aligning with established practices where expert review precedes controlled user trials [3].

### B. System Development

The dual-device system was developed iteratively using Unreal Engine 5.4 for PC components and Flutter for smartphone applications, with Android Studio as the primary development environment for mobile deployment. The PC component implements multiplayer networking through Unreal Engine's OnlineSubsystemSteam in a listen-server topology, where one player hosts the session and other players join as clients. Steam services provide session discovery, NAT traversal, player replication, and synchronized world state management. The mobile application handles learning task interaction including letter tracing, drawing activities, difficulty level selection, and color profile selection.

PC-mobile communication is implemented using Firebase Realtime Database as cloud-based middleware for task synchronization. The Unreal Engine game integrates the VaRest plugin to construct and transmit JSON-based control signals to Firebase via REST API endpoints. The mobile application listens to relevant session nodes and reacts to PC-initiated task triggers by opening corresponding tracing activities. Upon task completion, the mobile application updates session state in Firebase, which the PC game reads to trigger in-game events.

Voice communication is implemented using Steam Voice Chat accessed through the Steam overlay, with all players joining a shared voice channel independent of in-game spatial positioning. Development proceeded through prototyping, technical integration testing, and refinement cycles based on internal evaluation and expert feedback.

### C. Technical Evaluation

Technical evaluation assessed system stability, multiplayer synchronization, network performance, voice communication reliability, and smartphone-PC integration through iterative testing cycles conducted over three weeks under controlled laboratory conditions. The evaluation environment comprised mid-range PCs (Windows 11, Intel i5/AMD Ryzen 5, 8-16 GB RAM, integrated Intel Iris Xe and discrete GPUs), Android 12-14 smartphones (4-8 GB RAM), and local Wi-Fi (802.11ac) infrastructure with mixed wired and wireless PC connections. Multiplayer sessions were evaluated using 2-5 concurrent players in listen-server topology, with one PC acting as host and others joining as clients. Multiple test configurations examined sessions lasting 30-120 minutes, with each configuration tested repeatedly to assess consistency. Performance metrics included frame rate stability, network replication latency, bandwidth consumption, voice communication quality, PC-smartphone event triggering latency via Firebase, and session crash frequency. Testing was performed by adult developers across diverse hardware configurations to validate technical feasibility and identify performance bottlenecks prior to child user deployment.

### D. Expert Validation

Expert validation was conducted through iterative consultation with two consultant child and adolescent psychiatrists affiliated with a pediatric hospital, selected based on specialized expertise in developmental disorders, learning disabilities, and evidence-based educational interventions. The panel size reflects a documented national constraint: Sri Lanka has only 0.03 board-certified child and adolescent psychiatrists per 100,000 population, with only 12 certified nationally [22], making recruitment of such specialists a recognized limitation in Sri Lankan research contexts. While a broader expert panel would be preferable, the iterative multi-round consultation process partially compensates for panel size; an expanded panel including special educators and speech-language pathologists is planned for Phase 2. The validation process comprised multiple rounds of structured feedback sessions. In initial sessions, experts were provided with comprehensive gameplay demonstrations and detailed system walkthroughs explaining design objectives, target audience characteristics, cooperative gameplay structure, multisensory task integration, and emotional safety mechanisms. Following each session, preliminary feedback was incorporated into design refinements, after which subsequent validation sessions were conducted to review modifications and assess design maturity. Structured feedback focused on emotional safety mechanisms, developmental appropriateness for children aged 8-12 with



dyslexia, suitability of cooperative mechanics, alignment with evidence-based literacy intervention principles, potential implementation challenges, and feasibility for supervised educational contexts. Feedback was systematically collected through semi-structured discussions, documented with detailed notes, and synthesized for analysis (Table II), establishing design validation prior to child user studies. Iterative feedback yielded concrete revisions: voice-only communication replaced visible avatars to reduce social comparison; feedback language was revised to exploratory phrasing; and a 30-minute session ceiling was added per expert guidance. Both experts' ratings across all dimensions are reported as combined bars in figure 7 individual scores diverged by no more than 0.5 points on any dimension, indicating strong inter-rater agreement, and combined representation is therefore appropriate.

## V. RESULTS AND DISCUSSION

This section presents technical performance evaluation and expert validation findings, followed by discussion of system feasibility, pedagogical positioning, and comparison with existing dyslexia interventions.

### A. Technical Performance

TABLE I  
TECHNICAL PERFORMANCE METRICS

| Metric                               | Value           |
|--------------------------------------|-----------------|
| Frame Rate (Discrete/Integrated GPU) | 70-80/30-35 FPS |
| Network Latency (2/5 players)        | 60/150 ms       |
| Bandwidth/User (2/5 players)         | 140/450 kbps    |
| PC-Smartphone Event Latency          | 160-190 ms      |
| Task Delivery Accuracy               | >98%            |
| Session Stability (48h testing)      | 0 crashes       |

Technical evaluation across three weeks with 2-5 concurrent players demonstrated system feasibility. Table I summarizes the performance metrics, and the network scaling behavior is visualized in Figure 6.

Frame rates remained above playability thresholds across hardware configurations. Network latency scaled predictably from 60 ms (2 players) to 150 ms (5 players). PC-smartphone event delivery via Firebase achieved greater than 98% accuracy with mean latency 160-190 ms, acceptable for non-real-time literacy activities. The 1.7% failure rate resulted from Firebase timeouts under degraded network conditions; all events recovered automatically within 2-4 seconds via retry logic. Zero crashes across 48 testing hours confirmed architectural stability. The dual-device architecture demonstrates technical feasibility for supervised small-group contexts with 2-5 concurrent players, though scalability remains constrained by absence of large-scale field testing.

### B. Expert Validation Findings

Two consultant child and adolescent psychiatrists provided iterative feedback across multiple structured sessions. Table II summarizes the feedback across evaluation dimensions, and the corresponding ratings are presented in Figure 7.

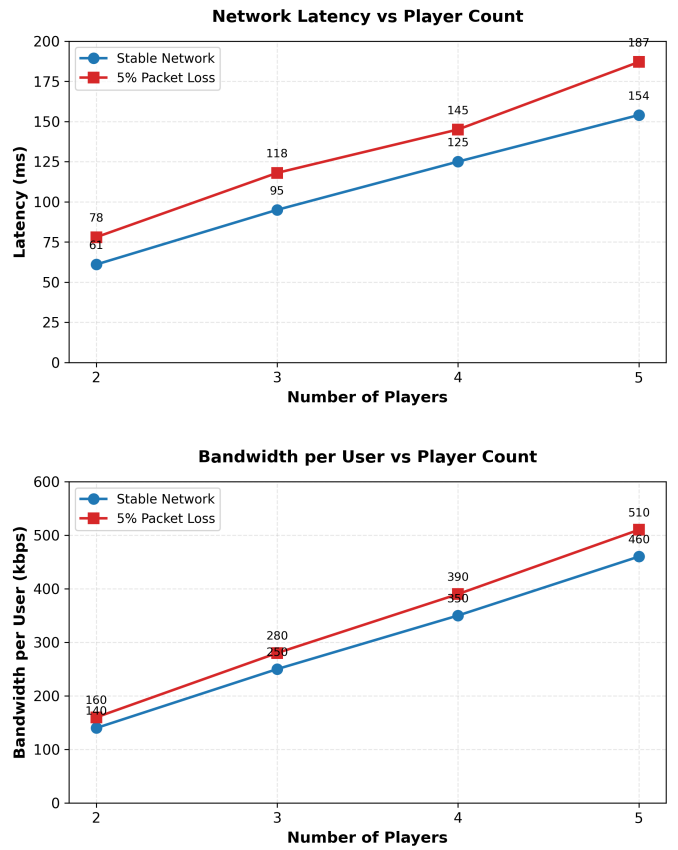


Fig. 6. Network latency and bandwidth scaling with player count

TABLE II  
EXPERT VALIDATION SUMMARY

| Dimension                | Expert Feedback                                                                 |
|--------------------------|---------------------------------------------------------------------------------|
| Emotional Safety         | Low-visibility interaction and non-punitive feedback reduce performance anxiety |
| Age Suitability          | Appropriate for 8-12 with facilitator support                                   |
| Visual Design            | Dyslexia-friendly colors, fonts, 3D letters viewed favorably                    |
| Cooperative Mechanics    | Shared goals without competition align with therapeutic principles              |
| Dual-Device Architecture | Balances engagement with private practice                                       |
| Multisensory Integration | Visual, auditory, tactile modalities align with evidence-based approaches       |
| Implementation           | Requires facilitator training and session duration limits                       |

Experts affirmed emotional safety mechanisms (low-visibility interaction, non-punitive feedback), dyslexia-friendly visual design, and cooperative mechanics as appropriate for the target population. The dual-device architecture was recognized as effectively addressing the tension between collaborative engagement and performance anxiety by enabling private literacy practice on smartphones while maintaining social connection through shared PC exploration. Multisensory task integration combining visual, auditory, and tactile modalities aligned with evidence-based dyslexia interventions [19]. Narrative scaffolding was identified as developmentally appropriate for ages 8-12. Constructive feedback emphasized facilitator training

requirements, session duration management for screen time concerns, and expanding task variety for sustained engagement.

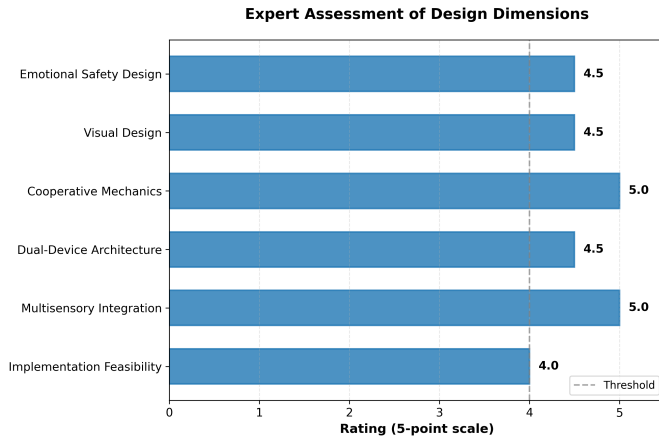


Fig. 7. Expert assessment ratings on a 5-point scale across six evaluation dimensions. Both experts provided identical or near-identical ratings on all dimensions; scores diverged by no more than 0.5 points on any single dimension, indicating strong inter-rater agreement

### C. Pedagogical Positioning

The Word Garden functions as a supplementary engagement tool for supervised contexts rather than replacing formal phonics instruction. The dual-device architecture addresses a fundamental tension by separating collaborative exploration (PC) from individual practice (smartphone), enabling private struggle with challenging tasks while maintaining social connection. Touch-enabled smartphones support kinesthetic learning [19], while PC provides immersive storytelling. The session-based connection mechanism reduces administrative overhead and privacy concerns. The following design principles are proposed for reuse by researchers and developers designing emotionally safe educational technology for children with learning difficulties: (1) dual-device cognitive compartmentalization, meaning private practice should be separated from shared social spaces architecturally; (2) cooperative-without-competition mechanics, meaning shared goals should be structured without leaderboards or rankings; (3) low-visibility interaction, meaning voice-only communication should be used to reduce social comparison; (4) non-punitive error framing, meaning incorrect attempts should be treated as exploratory iterations rather than failures; and (5) effort-based rewards, meaning participation and progress should be rewarded rather than comparative achievement.

### D. Comparison with Existing Solutions

Table III positions The Word Garden relative to established interventions. Digital tools such as GraphoGame, Lexia Core5, and Nessy Learning provide evidence-based phonics practice through adaptive single-user formats [11] without social interaction. The Word Garden employs dual-device cooperative multiplayer architecture, positioning social engagement and

emotional safety alongside literacy development. Platforms like Duolingo employ competitive mechanics including leaderboards [6], which may heighten anxiety for struggling readers [3]. The Word Garden's exclusion of competitive elements addresses performance anxiety, positioning it as complementary to validated phonics-based interventions.

TABLE III  
COMPARISON WITH EXISTING DYSLLEXIA INTERVENTIONS

| Feature      | GraphoGame | Lexia | Nessy | Word Garden |
|--------------|------------|-------|-------|-------------|
| Multiplayer  | ×          | ×     | ×     | ✓           |
| Cooperative  | ×          | ×     | ×     | ✓           |
| Dual-device  | ×          | ×     | ×     | ✓           |
| Voice comm.  | ×          | ×     | ×     | ✓           |
| Multisensory | ✓          | ✓     | ✓     | ✓           |
| Adaptive     | ✓          | ✓     | ✓     | Planned     |
| Evidence     | ✓          | ✓     | ✓     | Expert      |
| Competition  | ×          | ✓     | ✓     | ×           |

## VI. CONCLUSION

This paper presented The Word Garden, a cooperative multiplayer educational game investigating how dual-device architecture and cooperative mechanics might address gaps in dyslexia-focused digital interventions. The Word Garden integrates emotionally safe social interaction with literacy skill development through architectural separation of collaborative PC-based exploration from private smartphone-based literacy tasks, enabling children to engage in challenging literacy exercises without peer visibility while maintaining social connection through shared narrative progression and voice communication.

Expert validation by consultant child and adolescent psychiatrists confirmed alignment between design features and therapeutic principles supporting literacy intervention. The cooperative multiplayer structure, absence of competitive elements, and voice-only interaction appropriately address performance anxiety. Technical testing demonstrated system feasibility, including stable multiplayer synchronization, responsive smartphone-PC interaction, and acceptable performance with 2-5 concurrent players.

However, this study presents a formative evaluation without empirical assessment of learning effectiveness involving children. Expert validation reflects professional judgments based on system inspection rather than direct observation of child interaction. Technical evaluation was performed under controlled laboratory conditions, limiting conclusions regarding long-term field reliability. The system remains at prototype level, limiting insight into practical integration challenges. The prototype's restricted task range may limit sustained engagement, while reliance on stable connectivity may present challenges in settings with limited infrastructure. The expert panel comprised only two psychiatrists and testing was conducted by adult developers, limiting conclusions on real-world usability for the target population. Child user evaluation with an expanded expert panel, including special educators and

speech-language pathologists alongside psychiatrists, constitutes the immediate next phase of this longitudinal study. This Phase 2 evaluation will address empirical learning outcomes, usability with the target population, and real-world deployment feasibility, providing the empirical evidence base that the current formative phase intentionally defers.

This work establishes a foundation for investigating whether cooperative multiplayer environments integrating social support with literacy practice can effectively support both literacy development and social-emotional well-being in children with dyslexia. Future work will focus on controlled deployment with children, evaluating whether the cooperative structure achieves its intended goals, followed by longitudinal studies examining sustained engagement and broader deployment trials addressing scalability and integration within diverse educational contexts.

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